

PROGRAM 7 — Deploy Web App using Kubernetes & Minikube

STEP 1: Start Minikube

minikube start

STEP 2: Verify Minikube status

minikube status

STEP 3: Create Deployment YAML file

Create file:

nano nginx-deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
```

Save file.

STEP 4: Apply Deployment

kubectl apply -f nginx-deployment.yaml

STEP 5: Check Deployment & Pods

kubectl get deployments

kubectl get pods

STEP 6: Expose Deployment via Service (NodePort)

```
kubectl expose deployment nginx-deployment --type=NodePort --port=80
```

STEP 7: Check Services

```
kubectl get svc
```

STEP 8: Access the Web Application

```
minikube service nginx-deployment
```

This opens the Nginx welcome page in your browser.

STEP 9: Cleanup

```
kubectl delete service nginx-deployment
kubectl delete deployment nginx-deployment
```

OUTPUT:

Nginx deployed and accessed via Minikube service URL.

```

🌟 Enabled addons: default-storageclass, storage-provisioner

! /usr/bin/kubectl is version 1.28.15, which may have incompatibilities with Kubernetes 1.34.0.
  ■ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
🔧 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC:~/prog7$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC:~/prog7$ nano nginx-deployment.yaml
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC:~/prog7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC:~/prog7$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment  2/2     2             2           32s
test          1/1     1             1           10d
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC:~/prog7$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
kubia         1/1     Running   1 (107s ago)  3d23h
nginx-deployment-6f9664446b-bml6q  1/1     Running   0           52s
nginx-deployment-6f9664446b-rcf5d  1/1     Running   0           52s
test-6bb654b8f8-p82wd             1/1     Running   1 (3d14h ago)  3d23h
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC:~/prog7$ kubectl expose deployment nginx-deployment --type=NodePort --port=80
service/nginx-deployment exposed
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC:~/prog7$ kubectl get services
NAME          TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
kubernetes    ClusterIP     10.96.0.1      <none>          443/TCP           16d
kubia         NodePort      10.100.58.146  <none>          8080:31766/TCP   3d23h
kubia-http    LoadBalancer 10.108.190.36  <pending>       8080:30716/TCP   3d23h
nginx-deployment  NodePort      10.109.18.4    <none>          80:30109/TCP     11s
test          ClusterIP     10.104.246.143 <none>          80/TCP           10d
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC:~/prog7$ minikube service nginx-deployment
🔧 Opening service default/nginx-deployment in default browser...
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC:~/prog7$ Gtk-Message: 10:14:52.224: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:30109

